Assignment No. - 9

Title: CRUD Operations using MongoDB.

Problem Statement:

Design and implement basic Create, Read, Update, and Delete (CRUD) operations using MongoDB. Use the save method and logical operators where necessary.

Objective:

- To learn concept of NOSQL.
- To implement CRUD operation in MOngoDB.

Theory :

- Difference between SQL and NOSQL

	SQL	NoSQL
Database Type	Relational Databases	Non-relational Databases / Distributed Databases
Structure	Table-based	• Key-value pairs • Document-based • Graph databases • Wide-column stores
Scalability	Designed for scaling up vertically by upgrading one expensive custom-built hardware	Designed for scaling out horizontally by using shards to distribute load across multiple commodity (inexpensive) hardware
Strength	• Great for highly structured data and don't anticipate changes to the database structure • Working with complex queries and reports	• Pairs well with fast paced, agile development teams • Data consistency and integrity is not top priority • Expecting high transaction load

NOSQL

A NoSQL (originally referring to "non SQL" or "non relational") database provides a mechanism for storage and retrieval of data that is modeled in means other than the tabular relations used in relational databases. Such databases have existed since the late 1960s, but did not obtain the "NoSQL" moniker until a surge of popularity in the early twenty-first century, triggered by the needs of Web 2.0 companies such as Facebook, Google, and Amazon.com. NoSQL databases are increasingly used in bigdata and real-time web applications. [6] NoSQL systems are also sometimes called "Not only SQL" to emphasize that they may support SQL-like query languages.

Motivations for this approach include: simplicity of design, simpler "horizontal" scaling to clusters of machines (which is a problem for relational databases), [2] and finer

control over availability. The datastructures used by NoSQL databases (e.g. key-value, wide column, graph, or document) are different from those used by default in relational databases, making some operations faster in NoSQL. The particular suitability of a given NoSQL database depends on the problem it must solve. Sometimes thedata structures used by NoSQL databases are also viewed as "more flexible" than relational database tables.

Many NoSQL stores compromise consistency (in the sense of the CAP theorem) in favor of availability,partition tolerance, and speed. Barriers to the greater adoption of NoSQL stores include the use of low- level query languages (instead of SQL, for instance the lack of ability to perform ad-hoc joins across tables), lack of standardized interfaces, and huge previous investments in existing relational databases. Most NoSQL stores lack true ACID transactions, although a few databases, such as MarkLogic, Aerospike, FairCom c-treeACE, Google Spanner (though technically a NewSQL database), Symas LMDB, and OrientDB have made them central to their designs. (See ACID and join support.)

MongoDB

MongoDB is a cross-platform, document oriented database that provides, high performance, high availability, and easy scalability. MongoDB works on concept of collection and document. Database Database is a physical container for collections. Each database gets its own set of files on the filesystem. A single MongoDB server typically has multiple databases.

Collection

Collection is a group of MongoDB documents. It is the equivalent of an RDBMS table. A collection exists within a single database. Collections do not enforce a schema. Documents within a collection canhave different fields. Typically, all documents in a collection are of similar or related purpose.

Document

A document is a set of key-value pairs. Documents have dynamic schema. Dynamic schema means thatdocuments in the same collection do not need to have the same set of fields or structure, and common fields in a collection's documents may hold different types of data.

Sample Document

Following example shows the document structure of a blog site, which is simply a commaseparated key value pair.

```
{
_id: ObjectId(7df78ad8902c)title: 'MongoDB Overview', description: 'MongoDB is no sql
database',by: 'tutorials point', url:'http://www.tutorialspoi
nt.com',
 tags: ['mongodb', 'database', 'NoSQL'],likes: 100, comments: [ {user:'user1',
message: 'My first comment', dateCreated: new Date(2011,1,20,2,15),like: 0
},
{
user:'user2',
message: 'My second comments', dateCreated: new Date(2011,1,25,7,45),like: 5
}]
}
```

_id is a 12 bytes hexadecimal number which assures the uniqueness of every document. You can provide _id while inserting the document. If you don‟t provide then MongoDB provides a unique id for every document. These 12 bytes first 4 bytes for the current timestamp, next 3 bytes for machine id,next 2 bytes for process id of MongoDB server and remaining 3 bytes are simple incremental VALUE.

MongoDB ─ Advantages:
Any relational database has a typical schema design that shows number of tables and the relationshipbetween these tables. While in MongoDB, there is no concept of relationship.

Advantages of MongoDB over RDBMS

Schema less: MongoDB is a document database in which one collection holds different documents. Number of fields, content and size of the document can differfrom one document to another.
Structure of a single object is clear..0 No complex joins.
Deep query -ability.
MongoDB supports dynamic queries on documents using a document-based query language that'snearly as powerful as SQL.
Tuning.
Ease of scale -out: MongoDB is easy to scale.
Conversion/mapping of application objects to database objects not needed.
Uses internal memory for storing the (windowed) workin g set, enabling faster access of data.

Why Use MongoDB?
Document Oriented Storage: Data is stored in the form of JSON style documents. Index on any attribute
Replication and high availability Auto -sharding
Rich queries
Fast in -place updates Professional support by MongoDB
Where to Use MongoDB?
Big Data
Content Management and Delivery Mobile and Social Infrastructure User Data Management Data Hub

MongoDB Help
To get a list of commands, type db.help() in MongoDB client. This will give you a list of commands as shown in the following screenshot.

The use Command
MongoDB use DATABASE_NAME is used to create database. The command will create a new database if it doesn't exist, otherwise it will return the existing database.
Syntax
Basic syntax of use DATABASE statement is as follows:
use DATABASE_NAME

Example
If you want to create a database with name <mydb>, then use DATABASE statement would be asfollows:
>use mydb
switched to db mydb
To check your currently selected database, use the command db dbmb If you want to check your databases list, use the command show dbs.

>show dbs
local 0.78125
GBtest 0.23012 GB
Your created database (mydb) is not present in list. To display database, you need to insert at least onedocument into it.
>db.movie.insert({"name":"tutorials point"})
In MongoDB default database is test. If you didn't create any database, then collections will be storedin test database.
The dropDatabase() Method
MongoDB db.dropDatabase() command is used to drop a existing database.
Syntax
db.dropDatabase()
The createCollection() Method
MongoDB db.createCollection(name, options) is used to create collection. Syntax :
db.createCollection(name, options)
In the command, name is name of collection to be created. Options is a document and is used to specifyconfiguration of collection.

Parameter	Type	Description
Name	String	Name of the collection to be created
Options	Document	(Optional) Specify options about memorySize and indexing.

The drop() Method
MongoDB's db.collection.drop() is used to drop a collection from the database.
Syntax
db.COLLECTION_NAME.drop().
MongoDB supports many datatypes. Some of them are:
String: This is the most commonly used datatype to store the data. String in MongoDB must beUTF-8 valid.

Integer: This type is used to store a numerical value. Integer can be 32 bit or 64 bit depending upon
your server.
Boolean : This type is used to store a boolean (true/ false) value.
Double: This type is used to store floating point values.

Min/Max Keys: This type is used to compare a value against the lowest and highest BSONelements.

Arrays: This type is used to store arrays or list or multiple values into one key.
Timestamp: ctimestamp. This can be handy for recording when a document has been modified oradded.

Object: This datatype is used for embedded documents.
Null: This type is used to store a Null value.
Symbol: This datatype is used identically to a string; however, it's generally reserved for languagesthat use a specific symbol type.

Date: This datatype is used to store the current date or time in UNIX time format.
You can specify
your own date time by creating object of Date and passing day, month, year into it.

Object ID: This datatype is used to store the document‟s ID.
Binary data: This datatype is used to store binary data.

Code: This datatype is used to store JavaScript code into the document. Regular expression: This datatype is used to store regular expression. MongoDB ─ Insert Document

The insert() Method
To insert data into MongoDB collection, you need to use MongoDB's insert() orsave()method.

Syntax
The basic syntax of insert() command is as follows −
>db.COLLECTION_NAME.insert(document)
MongoDB ─ Query Document

The find() Method
To query data from MongoDB collection, you need to use MongoDB's find()method.Syntax
The basic syntax of find() method is as follows:
>db.COLLECTION_NAME.find()
find()method will display all the documents in a non-structured way.The pretty() Method To display the results in a formatted way, you can use pretty() method.

Syntax
>db.mycol.find().pretty()
Apart from find() method, there is findOne() method, that returns only one document.
RDBMS Where Clause Equivalents in MongoDB
To query the document on the basis of some condition, you can use following operation

Logical operators
AND in MongoDB Syntax
In the find() method, if you pass multiple keys by separating them by ',' then MongoDBtreats it as AND condition. Following is the basic syntax of AND −
>db.mycol.find({key1:value1, key2:value2}).pretty()

OR in MongoDB

Syntax

To query documents based on the OR condition, you need to use $or keyword. Followingis the basic syntax of OR −
>db.mycol.find({ $or: [{key1: value1}, {key2:value2}
] }).pretty()Using AND and OR Together
Example

The following example will show the documents that have likes greater than 100 and whose title is either 'MongoDB Overview' or by is 'tutorials point'. Equivalent SQL where clause is 'where
likes>10AND (by = 'tutorials point' OR title = 'MongoDB Overview')'
db.mycol.find({"likes": {$gt:10}, $or: [{"by": "tutorials point"}, {"title": "MongoDB Overview"
}]}).pretty()

{ "_id":
ObjectId(7df78ad8902c)
,"title": "MongoDB Overview",
"description": "MongoDB is no sql database","by": "tutorials point", "url":
"http://www.tutorialspoint.com", "tags": ["mongodb", "database",
"NoSQL"],"likes": "100" }

MongoDB's update()

MongoDB's update() and save() methods are used to update document into a

collection. The update()method updates the values in the existing document while the save() method replaces the existing document with the document passed in save() method.
MongoDB Update() Method

The update() method updates the values in the existing document.Syntax The basic syntax of update() method is as follows:

>db.COLLECTION_NAME.update(SELECTIOIN_CRITERIA, UPDATED_DATA)

MongoDB Save() Method

The save() method replaces the existing document with the new document passed in the save() method.Syntax
The basic syntax of MongoDB save() method is −

>db.COLLECTION_NAME.save({_id:ObjectId(),NEW_DATA}).

The remove() Method
MongoDB's remove() method is used to remove a document from the collection. remove() methodaccepts two parameters. One is deletion criteria and second is justOne flag.

deletion criteria: (Optional) deletion criteria according to documents will be removed.

justOne: (Optional) if set to true or 1, then remove only one document.Syntax
Basic syntax of remove() method is as follows:

```
>db.COLLECTION_NAME.remove(DELLETION_CRITTERIA)
```

Remove Only One
If there are multiple records and you want to delete only the first record, then set justOne parameter inremove() method.

```
>db.COLLECTION_NAME.remove(DELETION_CRITERIA,1)
```

Remove All Documents
If you don't specify deletion criteria, then MongoDB will delete whole documents from the collection.This is equivalent of SQL's truncate command.

```
>db.mycol.remove()
>db.mycol.find()
```

The Limit() Method

To limit the records in MongoDB, you need to use limit() method. The method accepts one numbertype argument, which is the number of documents that you want to be displayed.
Syntax
The basic syntax of limit() method is as follows:

```
>db.COLLECTION_NAME.find().limit(NUMBER)
```

MongoDB Skip() Method
Apart from limit() method, there is one more method skip() which also accepts number type argumentand is used to skip the number of documents.
Syntax
The basic syntax of skip() method is as follows:

```
db.COLLECTION_NAME.find().limit(NUMBER).skip(N UMBER)
```

The sort() Method

To sort documents in MongoDB, you need to use sort() method. The method accepts a document containing a list of fields along with their sorting order. To specify sorting order 1 and -1 are used. 1 isused for ascending order while -1 is used for descending order.
Syntax

The basic syntax of sort() method is as follows:

```
>db.COLLECTION_NAME.find().sort({KEY:1})
```

Installation steps:

->download the software “mongodb-linux-x86_64-3.4.9.tgz”
Copy to Download
[root@localhost Downloads]# tar -xvzf mongodb-linux- x86_64-3.4.9.tgz[root@localhost Downloads]# cd mongodb-linux-x86_64-3.4.9/ [root@localhost mongodb- linux-x86_64-3.4.9]# cd bin
[root@localhost bin]# ./mongod -dbpath /home/admin/
[root@localhost bin]# ./mongo

Installation steps in Windows:

Install MongoDB setup
Open "C:\Program Files (x86)\MongoDB\Server\3.0\bin"
mongod.exe ==> Server file ||mongo.exe ==> Client file
Create Folder like "D:\TE\data\db"
Run MongoDB Server
-> Open Command prompt as Administrator
-> cd C:\Program Files (x86)\MongoDB\Server\3.0\bin
-> mongod.exe --dbpath "D:\TEB\data\db"
Run MongoDB Client
-> Open Second Command prompt
-> cd C:\Program Files (x86)\MongoDB\Server\3.0\bin
-> mongo.exe

Conclusion : Thus we have studied the concept NOSQL-MongoDB and implemented.

FAQ :

What makes MongoDB the best?
If you remove an object attribute, is it deleted from the database? Explain with example.
How does MongoDB provide consistency?
Define MongoDB.
What are the key features of mongodb?
Which command is use to create database? Explain with example
Which command is use to drop database? Explain with example
What is the use of pretty() method? Explain with example
Which method is used to remove the document form the collection? Explain with example